

Programme	B. Sc. Computer Science				
Course Title	Security and Privacy in Cloud				
Type of Course	Elective				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	8. Basic understanding of computer networks, operating systems, databases, Cloud computing				
Course Summary	This course explores the security and privacy challenges in cloud computing environments. Students will learn about the fundamental principles, technologies, and best practices for ensuring the confidentiality, integrity, and availability of data in the cloud. The course also covers legal and ethical considerations related to privacy in cloud computing.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand fundamentals of security concepts (encryption, decryption)	U	C	Instructor-created exams / Quiz
CO2	Understand security design principles.	U	C	Assignment / Seminar presentations/ Exams
CO3	Analyze various threats to cloud security	U	P	Seminar Presentation / Group Tutorial Work/ Viva Voce
CO4	Understand various cloud security design patterns.	U	C	Instructor-created exams / Home Assignments
CO5	Explore various access control mechanisms and management schemes to ensure security in cloud.	U	P	Writing assignments/ Exams/ Seminar Presentations
CO6	Explore various levels of security in cloud infrastructure	U	F	Case Study/ Exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
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I	Fundamentals of Security in Cloud		14	22
	1	Overview of Cloud Security- Security services- Confidentiality, Integrity, Authentication, Non repudiation, Access control	2	
	2	Basics of Cryptography	2	
	3	Conventional and public key cryptography	4	
	4	Hash functions	2	
	5	Authentications	2	
	6	Digital Signature	2	
II	Security Design and Architecture for Cloud		12	18
	7	Security design principles for cloud computing- comprehensive data protection, end to end access control	2	
	8	Common attack vectors and threats	1	
	9	Network and storage- Secure Isolation strategies, Virtualisation strategies, inter- tenant network segmentation strategies, data protection strategies	3	
	10	Data retention, detection and archiving procedures for tenant data	2	
	11	Encryption, Redaction, Tokenisation, Obfuscation	2	
	12	PKI and key	2	
III	Access Control and Identity Management		12	18
	13	Access control requirements for Cloud infrastructure- user identification, authentication and authorization	2	
	14	Role based access control- multi-factor authentication, single Sign-on	2	
	15	Identity providers and service consumers	2	
	16	Storage and network access control options- OS Hardening and minimization	3	
	17	Intruder detection and prevention	3	
IV	Cloud Security Design patterns		10	12
	18	Introduction to design patterns	2	
	19	Cloud bursting	2	
	20	Geo-tagging	2	
	21	Secure cloud interfaces	2	
	22	Cloud resource access control	2	
V	Open Ended Module		12	
	1	Infrastructure security: Network level, host level, application level	4	
	2	Security management in the cloud	4	
	3	Audit and compliance	4	

References

1. "Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance" by Tim Mather, Subra Kumaraswamy, and Shahed Latif
2. "Cloud computing: Principles and Paradigms". Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Willey Publications

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	-	2	--	-	1	1						
CO 2	-	2	-	-	1	1						
CO 3	-	1	-	-	1	1						
CO 4	-	1	-	-	2	1						
CO 5	-	1	-	-	2	1						
CO 6	-	1	-	-	2	2						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	End Semester Examinations
CO 1	✓		✓
CO 2	✓		✓
CO 3	✓		✓
CO 4		✓	✓
CO 5		✓	✓
CO 6		✓	